SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA04II COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.*

QUESTIONS: 1 **Total Marks:100**

Annexure A

Technical Presentation

Code of Conduct:

I A. Gayathri / 192U26D342 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Gayathri

95/100

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Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Operating Systems in Banking Systems

CSA0411-OPRATING SYSTEM

Under the Guidance of
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Presented by
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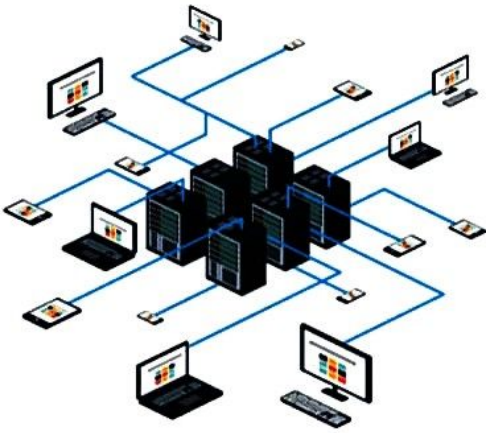
Introduction

- An Operating System (OS) manages computer hardware and software.
- Banking systems process thousands of transactions simultaneously.
- The OS manages CPU, memory, files, and network resources.
- It supports secure and reliable banking applications.
- It helps banking services run efficiently and continuously.



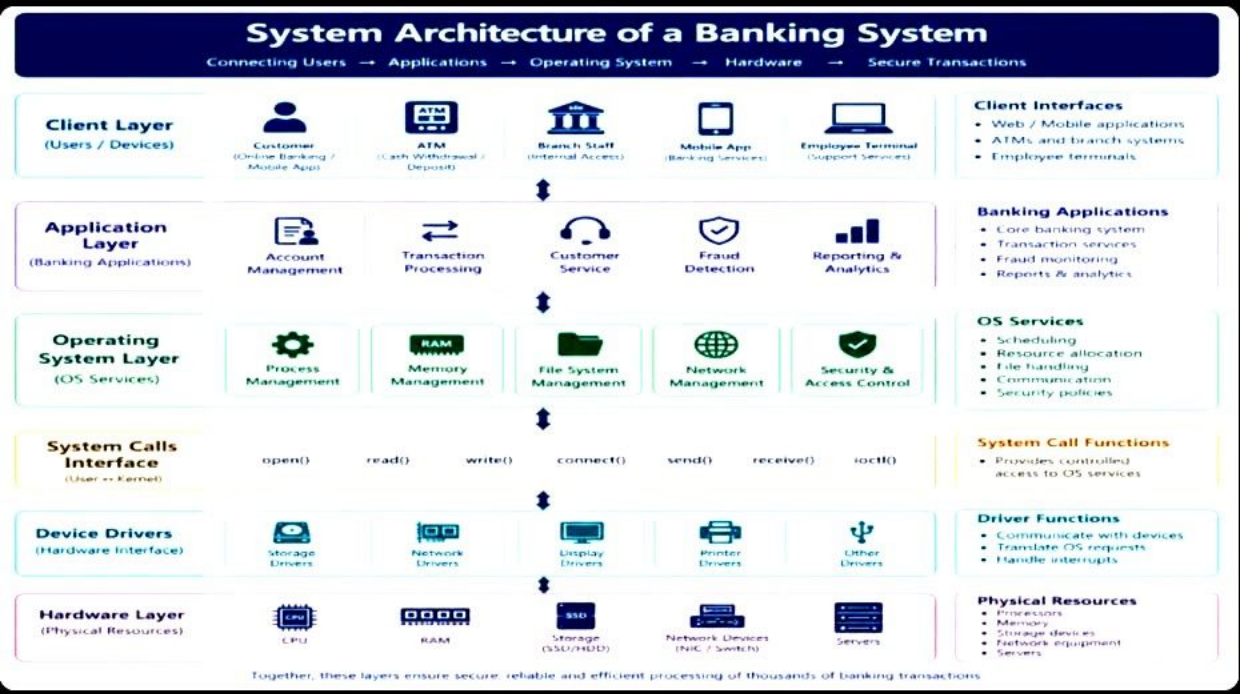
Operating Systems Suitable for Banking

- Multiuser OS: Supports many users and banking sessions.
- Multitasking OS: Handles multiple tasks at the same time.
- Multiprocessing OS: Uses multiple CPU cores for parallel processing.
- Network OS: Enables communication between connected banking systems.



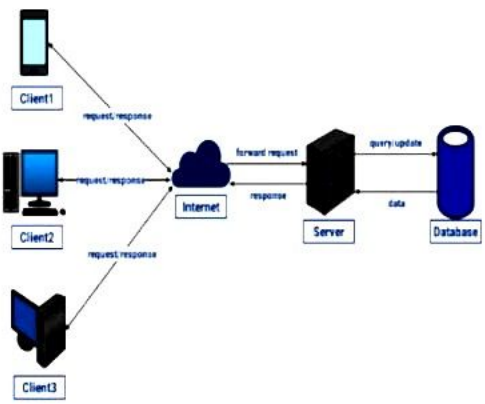
OS Structure Used in Banking

- User applications: Run banking and transaction software.
- System call interface: Connects applications to OS services.
- Kernel: Manages CPU scheduling, memory, and processes.
- Device drivers: Control storage and network hardware.
- Hardware: Provides CPU, RAM, storage, and network devices.



Role of System Calls

- File management: open(), read(), and write() access files.
- Process management: Creates and manages processes.
- Network communication: Sends and receives transaction data.
- Synchronization: Coordinates concurrent tasks and shared resources.
- Storage operations: fsync() requests that file data be flushed to storage.



System Calls in Transaction Processing

- Receive request: The banking application receives a transfer request.
- Validate request: Checks user permissions and transaction details.
- Process data: Uses CPU and memory to execute application logic.
- Store updates: Database operations save the transaction changes.
- Return result: The application confirms success or reports failure.

Example of a Banking Transaction

- Customer request: A customer transfers ₹500 to another account.
- Authentication: The system verifies the customer's identity.
- Balance check: The application checks the available balance.
- Atomic update: The database updates both accounts consistently.
- Confirmation: The customer receives a success message after the transaction commits.

Conclusion

- Suitable OS: Multiuser, multitasking, and multiprocessing capabilities are essential.
- OS structure: A modular kernel helps manage system resources.
- System calls: Provide controlled access to OS services.
- Security and reliability: Access controls, logging, backups, and monitoring protect banking operations.
- Final outcome: The system supports secure, accurate, and efficient transactions.

RUBRICS FOR CALCULATION

Technical Presentation					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness